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On the Diffusibility of High-Dimensional Latents

Reconstruction-Finetuned Encoders Collapse to Low-Dimensional Manifolds—Switch to Data Prediction to Fix Flow Matching

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Reconstruction-finetuned visual encoders exhibit a sharp collapse in effective dimensionality, leaving standard velocity prediction in flow matching to fit orthogonal noise directions outside the signal manifold; switching to $\mathbf{x}_0$ -prediction (clean data parameterization) focuses optimization on the true signal subspace and substantially improves generation quality in high-dimensional Representation Autoencoder latent spaces.

AT A GLANCE

Problem: Finetuning visual encoders for reconstruction collapses their effective dimensionality, making standard velocity prediction in flow matching inefficient because it must fit noise directions orthogonal to the signal manifold.

Why it matters: Using pretrained visual encoder feature spaces as diffusion latents is desirable for rich semantic representations, but generation quality is poor without addressing this geometric mismatch.

Method: Analyze effective dimensionality collapse in reconstruction-finetuned RAEs; show $\mathbf{x}_0$ -prediction avoids fitting orthogonal noise directions.

Result: Substantial improvement in FID and CLIP scores by switching parameterization without any other change.

Evidence: CLIP ViT-L/14 and DINOv2 RAEs, ECCV 2026 evaluation.

The Big Picture

Latent diffusion models (LDMs) achieve high-quality image generation by operating in the compressed latent space of a variational autoencoder (VAE) rather than pixel space. A natural extension is to use the feature spaces of powerful pretrained visual encoders—CLIP, DINOv2, and their variants—as the latent space, combining the generative power of diffusion with the rich semantic representations learned by these encoders.

Representation Autoencoders (RAEs) equip pretrained encoders with a learned decoder to enable image reconstruction from features. But many off-the-shelf encoders are trained for classification or contrastive objectives and discard fine-grained visual details. To enable high-quality reconstruction, RAEs fine-tune the encoder on reconstruction objectives.

This finetuning introduces a subtle but consequential geometric problem: while the nominal feature dimension remains large (768, 1024, or more), the learned features concentrate near a much lower-dimensional subspace. Standard flow matching with velocity prediction must fit noise directions orthogonal to this subspace, making optimization inefficient and generation quality poor. This paper diagnoses the problem and proposes a simple, principled fix.

Why Existing Approaches Fall Short

Standard VAE latents (Stable Diffusion) are designed for diffusion: their dimensionality is explicitly controlled and their geometry is well-conditioned. RAE latents inherit neither property by design.

Velocity prediction (v -prediction, ε -prediction) in flow matching assumes data fills the ambient space. When data concentrates on a low-dimensional manifold, the velocity field has large orthogonal components that carry no signal but must be learned, wasting capacity.

Higher nominal dimensionality makes this worse: more dimensions implies more orthogonal directions to fit, even though no additional signal is present.

Dimensionality reduction (PCA projection before diffusion) partially addresses the problem but requires separate engineering, discards representation structure, and introduces artifacts at reconstruction.

Core Mechanism

Effective Dimensionality Analysis. Feature dimensionality is quantified via the participation ratio of the feature covariance:

$$d_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}$$

where λ_i are the eigenvalues of the feature covariance matrix. RAE features show a sharp collapse: $d_{\text{eff}} \ll d_{\text{nominal}}$ —typically 5–15% of the nominal dimension after reconstruction finetuning.

$\mathbf{x}_0$ -Prediction. In flow matching with linear interpolation:

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

Velocity target: $\mathbf{v} = \mathbf{x}_0 - \varepsilon$, which has a large noise component $-\varepsilon$ that spans the full d -dimensional space,

forcing the model to fit orthogonal directions.

Data target: $\mathbf{x}_0$, which lies on the d_{eff} -dimensional signal manifold. Predicting $\mathbf{x}_0$ avoids the orthogonal noise component entirely.

Technical Formulation

Let $\mathbf{x}_0 \in \mathbb{R}^d$ be an RAE feature with $d_{\text{eff}} \ll d$, and $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.

The noisy sample at time t :

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \varepsilon$$

Velocity-prediction loss:

$$\mathcal{L}_v = \mathbb{E}_{t, \mathbf{x}_0, \varepsilon} [\|f_\theta(\mathbf{x}_t, t) - (\mathbf{x}_0 - \varepsilon)\|^2]$$

Data-prediction loss:

$$\mathcal{L}_{x_0} = \mathbb{E}_{t, \mathbf{x}_0, \varepsilon} [\|f_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

The velocity residual decomposes as:

$$\mathbf{x}_0 - \varepsilon = \underbrace{\mathbf{x}_0}_{\text{signal, on manifold}} - \underbrace{\varepsilon}_{\text{noise, spans } \mathbb{R}^d}$$

Fitting $\mathbf{x}_0$ only is equivalent to solving a projection onto the signal manifold, avoiding the orthogonal noise component. The optimization landscape for $\mathcal{L}_{x_0}$ is lower-dimensional and better-conditioned than for $\mathcal{L}_v$ when $d_{\text{eff}} \ll d$.

Learning or Inference Procedure

Training:

1. Finetune the pretrained visual encoder as a Representation Autoencoder (RAE) for reconstruction quality.
2. Extract RAE features for the training image set.
3. Compute d_{eff} to characterize the signal manifold.
4. Train a flow-matching generator with $\mathcal{L}_{x_0}$ instead of $\mathcal{L}_v$.

Inference:

1. Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.
2. Integrate the learned flow from $t = T$ to $t = 0$ using the $\mathbf{x}_0$ prediction.
3. Decode the generated feature $\mathbf{x}_0$ with the RAE decoder.

What the Guarantee Says

When $d_{\text{eff}} \ll d$, $\mathcal{L}_v$ is provably harder to optimize than $\mathcal{L}_{x_0}$ because the orthogonal noise components constitute a high-dimensional regression problem with no signal—they

can only be fitted by memorization. The data parameterization eliminates this by construction. The benefit scales with d/d_{eff} : the greater the effective dimensionality collapse, the larger the expected improvement.

Experimental Findings

- Evaluated with CLIP ViT-L/14 and DINOv2 RAEs at feature dimensions 768 and 1024
- d_{eff} is 5–15% of d_{nominal} after reconstruction finetuning
- $\mathbf{x}_0$ -prediction achieves substantially better FID compared to velocity prediction in the same RAE latent space
- CLIP similarity scores also improve, confirming benefit beyond distributional statistics
- Results are robust across encoder types and RAE decoder architectures
- FID gap between parameterizations correlates with d/d_{eff} , directly supporting the manifold mismatch hypothesis

Ablations and Interpretation

No RAE finetuning (frozen encoder): When the encoder is not finetuned for reconstruction, d_{eff} is higher and the gap between parameterizations is smaller, confirming the mechanism.

Alternative fix (PCA projection): Projecting to the principal subspace before velocity prediction partially recovers quality but is less elegant: it requires separate PCA fitting, discards features, and underperforms $\mathbf{x}_0$ -prediction.

Time t dependence: The difficulty of velocity prediction is greatest at intermediate t values where ε and $\mathbf{x}_0$ contribute equally; $\mathbf{x}_0$ -prediction remains well-conditioned throughout.

Nominal vs. effective dimensionality: Higher nominal dimension is strictly worse under velocity prediction but not under $\mathbf{x}_0$ -prediction, removing the disincentive to use richer representations.

Reference

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